

2nd International Conference on System-integrated Intelligence SysInt2014:
New Challenges for Product and Production Engineering

A Multi-Disciplinary Modeling Technique for Requirements Management in Mechatronic Systems Engineering

Georg Hackenberg^{*a}, Christoph Richter^b, Michael F. Zäh^b

^aChair IV: Software & Systems Engineering, Technische Universität München,

^bInstitute for Machine Tools and Industrial Management (iwb), Technische Universität München.

Abstract

Machines and plants continuously increase in complexity due to higher customer expectations regarding their purpose and flexibility of use. For this reason, machines and plants development processes need to adapt continuously to the present situation. In addition, the growing meaning of electronic and software components in machines and plants complicates the engineering since the interaction of different disciplines leads to increasing coordination efforts and problems regarding the interface between these disciplines. To face these challenges, the suitable handling of arising requirements is one key factor for the success of engineering processes. Therefore, a multi-disciplinary modeling technique for requirements management is presented in this paper. After a summary of approaches for model-based development and requirements management in the context of mechatronic systems engineering, the modeling technique with its concepts, terms, tasks and workflows is illustrated. To demonstrate the feasibility of the proposed approach, the modeling technique was applied to a miniaturized packaging plant, which is currently developed at the Institute for Machine Tools and Industrial Management (iwb) of Technische Universität München for teaching students and benchmarking engineering methods and tools. Finally, an outlook with extensions of the concept for fault-tolerant systems and the integration of further engineering tools in order to promote a consistent engineering tool-chain is given.

© 2014 Published by Elsevier Ltd. Selection and/or peer-review under responsibility of the Organizing Committee of the SysInt 2014

Keywords: Requirements Management, Mechatronics, Interdisciplinary Development, Systems Engineering

* Corresponding author. Tel.: +49-89-289-17080 ; fax: +49-89-289-17307.

E-mail address: hackenbe@in.tum.de.

1. Introduction

The machine and plant engineering is currently subjected to global mega-trends like a growing customer-individuality in products or a shortening of innovation cycles [1]. Besides technological improvements or the optimization of production, development processes of machines or plants form a key factor to meet current and future challenges [2]. Since software is already the main innovation driver in many industry branches [3], the still mechanically dominated engineering principles have to be substituted by mechatronic development approaches, which allow an interdisciplinary engineering [4]. Especially the interaction of different disciplines (e.g. mechanics, electrics and software) leads to various problems for enterprises, which can be reflected in a poor product quality or delays in the commissioning [5].

To overcome this situation, the model-based mechatronic development of machines and plants is one common approach [4]. In contrast to the application of single engineering models (e.g. from CAD-tools), model-based mechatronic development means the alignment of the whole engineering process with an interdisciplinary model of the system [6]. Through the development of such models, engineers from all involved disciplines gain a common understanding of the mechatronic system and are able to contribute their competencies and expertise [4]. In addition, the application of a model-based mechatronic development approach allows the integration of further development tasks, like for example requirements management. Especially in a machine or plant development process, where requirements are usually handled with documents, this integration has an enormous potential to enhance the requirements management [7]. This in turn is necessary, since the fulfillment of customer requirements in time and on budget is a decisive factor to stay successful on the market [8].

For this reason, the paper presents a multi-disciplinary modeling technique that allows a modeling and evaluation of requirements throughout the engineering process. Therefore, the paper is structured as follows: In Section 2 the state of the art is briefly reviewed. On this basis, the modeling technique is outlined. Therein the method of how requirements can be modeled and tested using simulation is discussed (Section 3). Subsequently, Section 4 illustrates the application and evaluation of the modeling technique using an academic case study. Finally, the paper finishes with a brief summary and an outlook to further activities like the extension of the modeling technique to fault-tolerant systems.

2. State of the Art

In the following subsections current approaches in the fields of *model-based development* and *requirements management* – especially in the context of mechatronic systems engineering – are summarized in order to indicate deficiencies and derive potentials for advances.

2.1. Model-Based Development in Mechatronic Systems Engineering

Model-based development principles originated in the field of software engineering. Especially object-oriented modeling tools and the related “Unified Modeling Language” [9] contributed to a broad acceptance [10]. In general, the objective of model-based or model-driven software development is to supply the engineering process by the targeted use of models, which allow an abstract and easy-to-understand description of software and an automatic generation of the respective source code [10].

Whilst the model-based development is already well established in software engineering, the application within the field of mechatronic systems is still in an initial stage [11]. The main objective is similar to the one from software engineering. By using interdisciplinary and abstract models of a mechatronic system, the collaboration of all involved engineers is simplified since a common

understanding of the system to be developed is generated [6]. For this purpose, various approaches have been developed in the scientific environment, which are briefly presented in the following.

A first step towards an interdisciplinary mechatronic development is a consistent and abstract description of the system to be developed. Therefore, numerous modeling languages like the SysML [12] have been developed up to today. Particularly, function-oriented methods including the example from Kallmeyer [13], which deal with the depiction of system functions, are getting more attention as they permit a solution-neutral modeling. Such modeling languages form the basis for an integrated model-based development. Eigner et al. [4] extend the well-known V-Model from [14] in all development phases with model-based development methods using the mentioned language SysML. Especially the systems specification and analysis are focused to allow early virtual tests of the respective product. A comparable approach was focused within the research projects “Födera” [15] and “AQUIMO” [16]. While the first research project dealt with the set-up of an information architecture for mechatronic system development, in “AQUIMO” an engineering tool was developed, which allows a formal modeling of interdisciplinary information. In addition, a process model was outlined, which describes how the engineering tool is used throughout a development process. Another approach for an easy modeling of mechatronic systems was illustrated by Gausemeier et al. [17]. The specification technique CONSENS allows an interdisciplinary description of mechatronic systems using eight different views on a mechatronic system also including a view for requirements and use-cases. In addition, there exist numerous approaches that also contribute to a model-based development of mechatronic systems. Examples include the approach from Thramboulidis [18], who introduced different abstraction layers for mechatronic systems, or from Lüder et al. [19], who outlined a mechatronical modularization of manufacturing systems in order to allow an easy modeling and system analysis.

2.2. Requirements Management in Mechatronic Systems Engineering

Like model-based development, also requirements management has its origin in software engineering [20]. In this field, various approaches were developed during the last years in order to enable a purposeful handling of requirements. Davis [21], for example, defines four important subsets of requirements management activities: requirements elicitation, triage, specification and change management. In addition, he suggests different methods and techniques for each subset (e.g. user interviews for the elicitation) [21]. Pohl & Rupp [22] also propose various methods for the requirements elicitation, but focus especially on the requirements documentation distinguishing between natural-language and model-based documentation. Besides these approaches also own procedure models have been developed including the model from Hood & Wiebel [23], who define an iterative cycle of elicitation, specification, analysis and review for the definition of requirements.

In the approaches presented above the requirements management is defined as a stand-alone development activity. However, the multi-disciplinary nature of mechatronic systems necessitates a requirements management integrated in the systems development process [20]. Regarding this field, especially the interface between requirements management and systems engineering was considered in detail, like for example by Hood et al. [24]. But even mechatronic systems have been focused including the approach from Schedl [20]. She defines a three-phase procedure consisting of an elicitation planning phase, a description phase and a requirements usage phase throughout the development process. In addition, Jung [25] focuses especially on the requirements analysis in interdisciplinary development environments using an iterative and development oriented method. Further examples include the approach from Reiss & Schuller [26], who propose a workflow-based approach for requirements engineering, or from Gürtler et al. [27], who integrate the requirements analysis in the system conception phase using the example of a product-service-system.

The preceding section shows that various efforts have been made regarding the requirements management in mechatronic development processes. However, these approaches mainly describe the requirements elicitation and handling, but do not focus on the integration into the systems development. This in turn, especially combined with a model-based development, could lead to further advantages like for example an automatic requirements verification using system simulation. For this reason, a modeling technique for requirements management included in the systems development is outlined in the following.

3. Integrated Mechatronics Systems and Requirements Modeling Technique

The modeling technique presented in the following was originally developed for the model-based development of software for embedded systems [28]. This technique was first extended to mechatronic systems by integrating methods for graphical editing of physical components and a simulation environment for space-intensive systems [29]. Besides enabling an interdisciplinary system development, one main objective was to build up behavior-oriented spatio-temporal engineering models, which allow a virtual commissioning of the associated system control [5]. In the research project “IMoMeSA” this modeling technique is currently extended to enable an advanced model-based development of mechatronic systems. Besides the integration of requirements management, the research tasks focus on the automatic code generation with fault-handling and the modularization of mechatronic systems.

The basic structure of a modeled mechatronic system using the proposed technique is displayed in Figure 1. Mechatronic components with defined interfaces form the central modeling element for the description of machines and plants. Internally, each component can be displayed in four modeling views, which allow an encapsulated presentation of several model contents. First, *scenarios* in form of message sequence charts can be mapped to each component in order to define test cases for the verification of a mechatronic component or the entire system. In addition, *functions* can be added in terms of state machines to define sequences of activities the later implementation has to fulfill. These functions are essential for deciding which (mechatronic or discipline-specific) subcomponents are planned within the *decomposition* modeling view. Each mechatronic subcomponent also contains the four modeling views creating a hierarchical mechatronic structure of the system to be developed. Finally, each component can be seen from a *geometries* modeling view, where material interfaces or the physical representation of a component can be sketched.

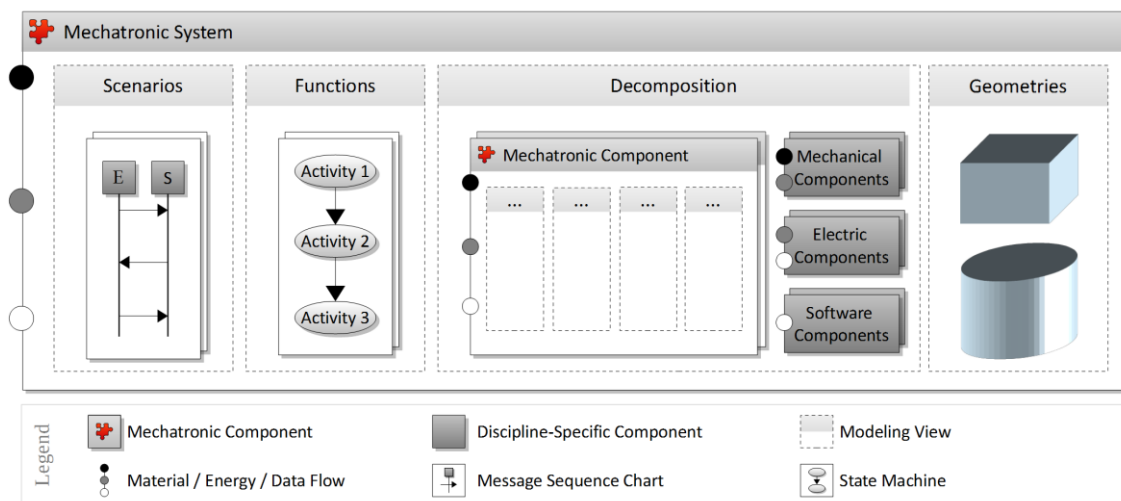


Figure 1. Basic structure of a modeled mechatronic system

For the modeling of requirements no separate modeling view is provided since scenarios and functions already describe requirements regarding the system behavior. In addition, constraints can be modeled and mapped to components, scenarios or functions. How this integration of the requirements management in the system conceptualization phase is carried out in detail, is described in the following subsections.

3.1. Concepts

One main objective during the set-up of the presented technique was to keep the number of modeling concepts to a minimum. Hence, an easy to understand modeling is realizable, especially in the early development. Within the four modeling views from Figure 1 different *concepts* are applied, which are defined in the following:

- *Components* are the key element within the presented modeling technique. They represent containers for all other modeling concepts irrespective of whether the content is mechatronic or discipline-specific. This simplifies the set-up of a modular structure as the content of components to be developed usually cannot be specified in advance. In addition, *templates* exist, which differ from components only as they may be used several times within one mechatronic system using instantiation.
- To define the interfaces of the components *material*, *energy* and *data ports* are used. In addition to the further distinction in *input* and *output* ports, energy and data ports are typically described by a specific type (e.g. kinetic vs. electric energy, Boolean vs. Integer). On the other hand, material ports are characterized by a geometric representation using the geometries modeling view. Thus volumes can be defined where materials can be transferred between components. The connection of input and output ports is implemented by *channels*, which define a source and target of a material, energy or data flow.
- *Processes* form a third central modeling element and are used for both scenarios and functions. The distinction results from the use and visual presentation of this concept (see terms). Generally, processes are characterized by *states* and *transitions*, the latter defining changes between various states. Further, transitions contain *guards*, which describe port conditions that have to be satisfied, and *actions*, which set ports using *expressions*. The expressions range from simple value assignments to complex three-dimensional vector descriptions for kinetic energy ports.
- Alongside the functional requirements, which result from scenario and function descriptions using the processes concept, *constraints* can be modeled. Generally, constraints consist of a *description* and a Boolean *expression* that makes qualitative or quantitative statements about mechatronic components. Besides numeric constraints (e.g. energy consumption), also constraints regarding geometric targets (e.g. operating dimensions) can be considered using the geometries modeling view. Concerning the allocation of constraints within the modeling technique, it is defined that they can be mapped either to components or to states in the scenario or function modeling view. Using this allocation, also constraints can be modeled, which are only valid in specific states (e.g. dwell times).
- Finally, the definition of geometric *parts* is supported by the proposed modeling technique. In contrast to material flow ports and geometric constraints, parts are physical representations of components and therefore part of the subsequent implementation. Therefore, only base geometries are applied since the main modeling objective is a common functional understanding among participating disciplines.

As mentioned above, four *modeling views* are applied, which allow an encapsulated presentation of several model contents. On the one hand, this means that all geometries of a component (material flow ports, geometric constraints and parts) are displayed and edited in one specific view (3D-editor). On the other hand, it means that scenarios are represented by message sequence charts and functions by state machines, even though both base on the process modeling concept.

3.2. Terms

In order to ensure the definiteness of the presented approach, a unique terminology is necessary. The terms that are needed to describe several contents of the modeling technique, in addition to the terms resulting from the concept, are defined in the following:

- Components can be divided in *composite* and *atomic* components. While an atomic component directly implements the required functionality, a composite component applies subcomponents for this implementation.
- *Scenarios* describe sequences of activities of a component's environment. Therefore, input ports of the component under consideration are written in the process definition (using actions) and it is checked whether the component reacts correctly regarding the output ports (using guards). These sequences can be visualized easily using message sequence charts with the entities *environment* and *component*.
- On the other hand, *functions* are used to determine a component's behavior from the component's points of view. Therefore, possible states are initially identified and modeled using the process concept. A change of these states depends on the component's input ports (using guards), whereby also possible output ports can be written (using actions). These closed-loop processes can be visualized easily using state machines.
- Functions can be further classified in *declarative* and *imperative* functions depending on whether or not actions are applied. This distinction is important as imperative functions can only be applied to atomic components and declarative function to composite components in order to provide a system simulation.

3.3. Tasks

As mentioned above, the technique allows to model mechatronic systems using mechatronic components and their modeling views. Yet, the model is only the first step to guarantee a purposeful development of mechatronic systems. Another important task is the simulation of the system model in order to evaluate system functions or determine the fulfillment of requirements. A third task results from the fact, that modeling in the proposed technique is only possible up to a certain level of detail since the aim of the model is a common functional understanding of the system to be developed. For this reason, all partial results have to be refined subsequently using discipline-specific engineering tools, like for example CAD-tools for purely mechanical design. Figure 2 illustrates the three identified tasks and shows how they are carried out in sequence.

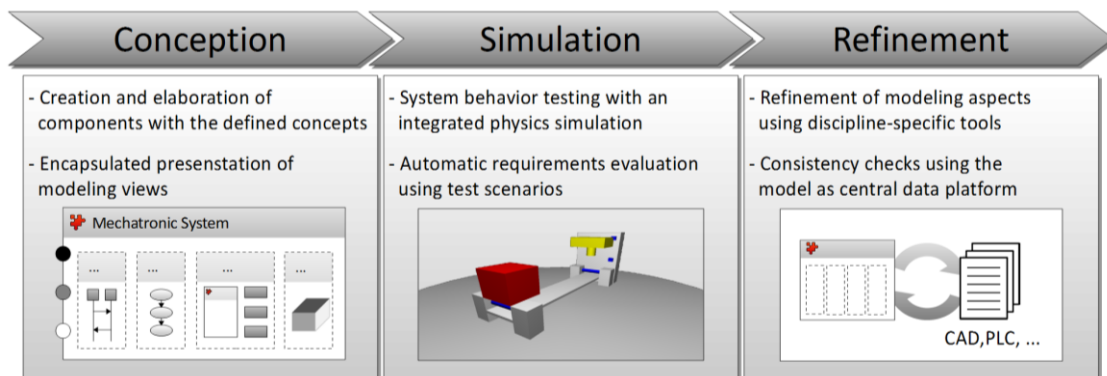


Figure 2. Sequence of engineering tasks for mechatronic systems development

The first task for mechatronic systems development deals with the *conception* of the system to be developed. In this phase the basic system structure and behavior are set by creating and elaborating mechatronic components using the already introduced modeling concepts and views.

The second task addresses the *simulation* in order to evaluate system behavior and enable rich communication among participating disciplines using the visualization of the simulation. For the fulfillment of this task, the mechatronic system model is converted to an executable test suite. Thereby, one test case is built for each scenario. Internally, the test cases run their respective scenario process as well as the imperative and declarative function processes of the system in combination with an integrated physics simulation of rigid body dynamics until the final state of the scenario process is reached. Hereby, imperative functions and the physics simulation determine the actual system behavior, while declarative functions serve as behavior monitors. Technically, the imperative function processes and the physics simulation are coupled through material and kinetic energy ports. Finally, modeled constraints are checked continuously during process execution and exceptions are thrown upon their violation. The test cases report a success or a failure depending on whether or not exceptions occurred during execution.

As mentioned above, modeling within the proposed technique is only possible up to a certain level of detail (e.g. base geometries within the mechanical design). For this reason, the modeling technique finally allows a *refinement* of development results using further engineering tools. This refinement is carried out by automatic transformations of model contents to discipline specific tools using standard data exchange formats (e.g. XML). In addition, methods are currently researched, which allow a re-integration of discipline-specific results in order to save the consistency throughout the entire development process.

3.4. Workflows

Since the defined modeling tasks cannot be executed sequentially, a workflow is defined that describes in which order mechatronic systems can be modeled using the proposed technique. The workflow presented in Figure 3 illustrates the basic information flows between the defined modeling tasks starting with a product idea and resulting in a virtually commissioned prototype. In addition, the conception task is displayed in detail in order to explain how the four defined modeling views can be elaborated.

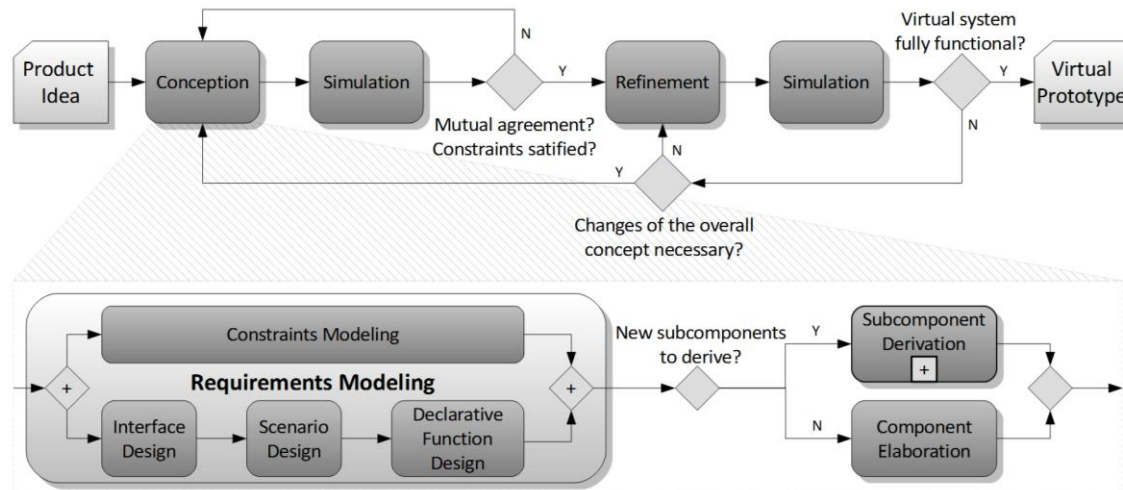


Figure 3. Workflows

The whole development process starts with a product idea, which emerges from market observations or customer orders. This idea is first transferred in a system concept using the modeling concepts and views. Subsequently, the modeled system can be simulated in order to determine whether all constraints are satisfied in the defined scenarios and a mutual agreement among involved disciplines is reached. If one of these conditions is not satisfied, the system conception has to be reviewed typically resulting in an iterative procedure between the conception and simulation tasks. Once the system is completely conceptualized, the interdisciplinary system model has to be refined using discipline-specific engineering tools and methods. These concept expansions have to be transferred back to the model as they may affect other disciplines and, thus, the required system behavior. Furthermore, this re-integration enables the repeated use of the simulation in order to check whether the refined system still satisfies the required system behavior and constraints. If the simulation shows that the system is not completely functional, it has to be decided whether only the refinement process or the entire system concept has to be reviewed. If, on the other hand, the refined concept still satisfies the required system behavior, the virtual system development is finalized resulting in a virtually commissioned prototype.

Since the conception task is the most important one regarding the requirements management, the belonging sub-workflow is also displayed in Figure 3. At the beginning, the interface of the system to be developed is defined using ports. On this basis, scenarios and declarative functions can be specified. In parallel with these activities, constraints can be modeled and assigned to the system or its scenarios and functions. These activities and the respective concepts essentially form the presented requirements modeling approach. Subsequently, the involved engineers have to decide whether or not subcomponents should be derived in order to decompose the required functionality. If it is decided to apply new subcomponents, those can be created using the decomposition modeling view. At this point, the activities described above (plus the connection of derived subcomponents with channels) are repeated resulting in a recursive modeling procedure. This recursion does not end until atomic components are derived exclusively. These atomic components can then be elaborated using imperative functions (necessary) and parts (optionally) in order to allow an executable simulation. Current research activities deal with the enhancement of this module elaboration (e.g. simple circuit diagrams for electronic components) and the integration of a mechatronic modular system for the easy re-use of already established mechatronic components during subcomponent derivation.

4. Case Study

In the following, the presented modeling technique is evaluated using an academic case study. After an introduction of the use-case, the conception and simulation tasks are outlined until finally the proposed approach is discussed.

4.1. Introduction

For benchmarking various mechatronic development methods and teaching students mechatronic principles, a miniaturized production plant is currently set up at the Institute for Machine Tools and Industrial Management (*iwb*). The main function of this plant is to transform paper into stamped boxes, which are filled with desired goods. For the fulfillment of this function, the plant is decomposed in functional components like for slicing, stamping, folding and gluing. In this paper, the focus is on the stamping component, which receives a sliced paper, prints a stamp onto it and delivers the stamped paper at a specific end location within a limited amount of time and energy consumption. In the following, the developed model of the stamping component (and derived subcomponents) is described with the focus on the requirements modeling (i.e. scenarios, declarative functions and constraints).

4.2. Conception

According to the defined workflows, the conception starts with the definition of an interface, which consists of a material input and output port (“start” / “end”). Based on this interface, a scenario was defined generating a new paper at the material input port and waiting for the stamped paper to appear at the material output port within a certain amount of time (Figure 1b).

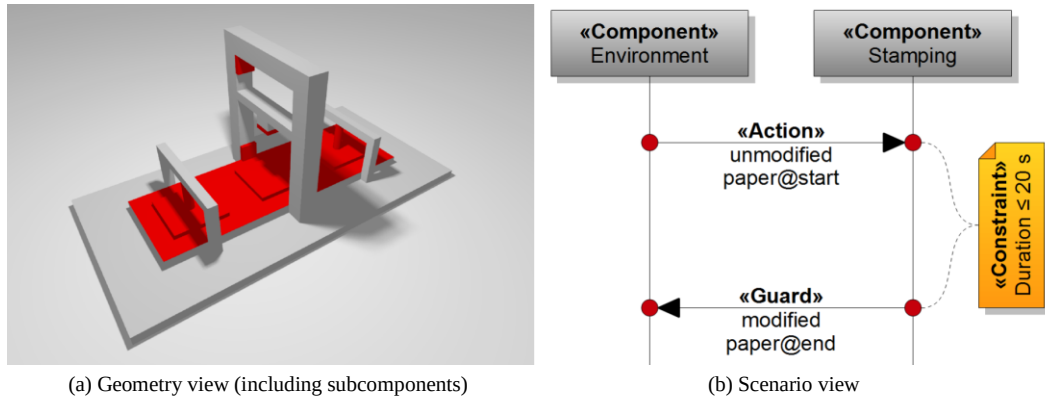


Figure 4. Geometry view and scenario specification of the *stamping component*

From this scenario a declarative function was derived switching between waiting and working states (Figure 5a). The stamping component switches from waiting to working state in case a material is observed at the material input port. In contrast, the stamping component switches from working to waiting state in case a stamped paper is observed at the material output port. In addition, a duration and an energy consumption are attached to the working state.

In this study the declarative function was considered to be too complex to be implemented directly. Thus, a decomposition in two subcomponents was defined: A transportation component and a manipulation component, which are connected through material and data ports respectively channels (Figure 5b). The decomposition resembles the two independent tasks of paper transportation (from “start” to “end”) and paper manipulation (from unmodified to modified). Since one decomposition step is sufficient for the understanding, only the transportation component is described in the following.

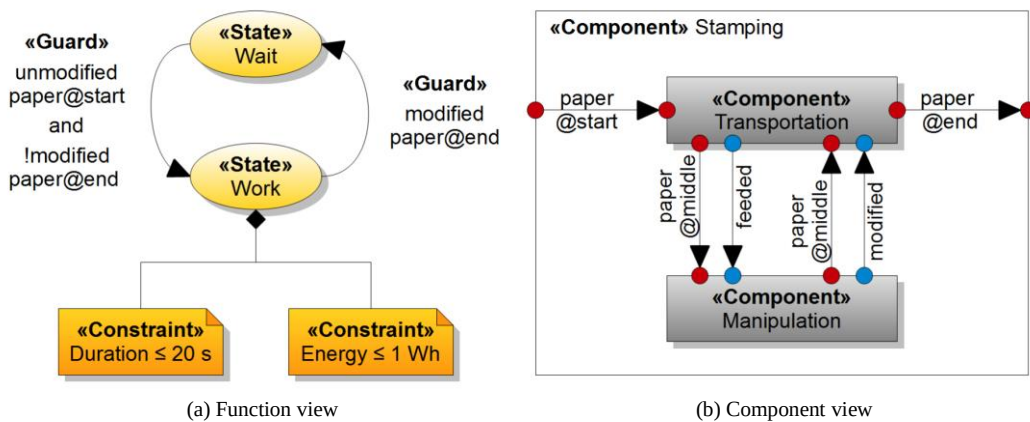


Figure 5. Declarative function and decomposition specification of the *stamping component*

In analogy to the transportation component, the interface consisting of material and data ports was defined first. While the two material ports from the stamping component could be adopted, a third material output port at the manipulation component location was defined (squares in Figure 6a). In addition, two data ports were defined; one output port to tell the manipulation component that material is available, one input port to receive the information that the material manipulation is finished.

Based on the interface, a declarative function was specified (Figure 6b). The component starts in waiting mode. As soon as unmodified material is observed at the start location, it is switched to the feeding state until unmodified material is observed at the manipulation location. Then, the transportation component waits again until modified paper is observed and the related signal at the data input port is received. Finally, the transportation component switches to removal state until modified material is observed at the end location. Further, duration and energy constraints were defined for the feed and remove states distributing equal portions of available resources.

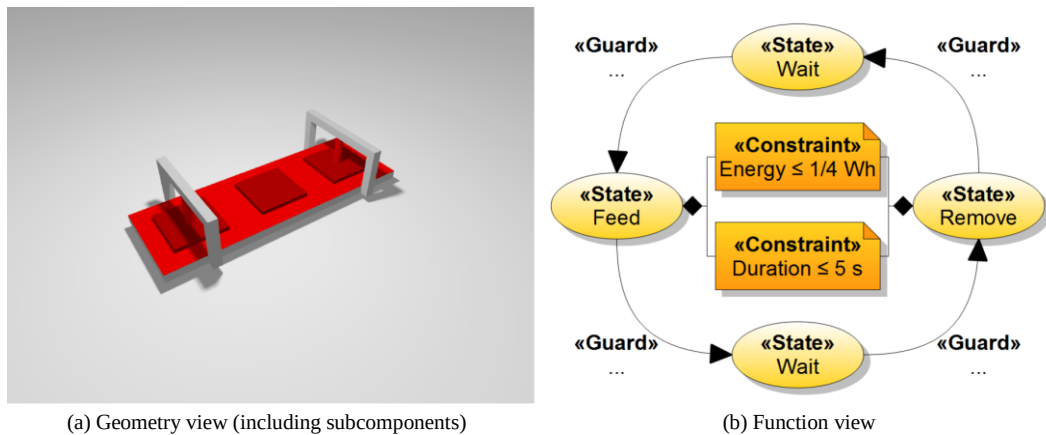


Figure 6. Geometry view and declarative function specification of the *transportation component*

At this point, the declarative function was analyzed again considering a further decomposition. Although a third and final level of decomposition was applied, a detailed presentation of derived subcomponents is omitted here. It is sufficient to point out that these actuators, sensors and other necessary components (e.g. a control program) were modeled with imperative functions and parts in order to enable a simulation of the modeled system.

4.3. Simulation

Once all imperative functions of the atomic components at the third decomposition level are defined, the model of the stamping component can be simulated as described in section 3.3. Therefore, the model is converted to the applied test suite (i.e. JUnit framework [30]). There a test case is built for the defined scenario (Figure 4b), which is automatically executed. During the simulation, first a paper is generated at the material input port, which is detected by a modeled light sensor. Subsequently the modeled control program starts the engine of a conveyor belt, which was introduced at the third decomposition level to fulfill the transportation task. This conveyor belt transfers energy to the paper resulting in a translational motion of the paper. Using further atomic components (with imperative functions) the entire behavior of the stamping component can be simulated analogously. The integration of a physics engine (i.e. JBullet framework [31]) enables a continuous visualization of motions during the simulation. Besides this

visualization, the modeled constraints are checked automatically. In the case study different conveyor belt engines were applied resulting in different velocities of the paper. Thus it could be shown that the modeled time constraints were violated in dependence of the applied engine.

4.4. Discussion

The case study showed how the proposed modeling technique enables seamless requirements management across discipline borders in the mechatronic engineering process. In particular, the technique allows to constrain the solution space gradually through successive refinement, while leaving the actual solution for the implementation phase. The refinements shown in the case study were carried out using subcomponents from the overall system, therefore describing a mechatronic decomposition of the overall engineering problem. Consequently, also the responsibilities of the individual disciplines for providing certain functionality are left open fostering the mechatronic engineering principle. One major benefit of decomposition and requirements modeling is that different implementations (e.g. mechatronic components from different vendors) can be plugged in as long as they are conforming to the requirements model. Consequently, both requirements models and implementation models can be reused across different projects. Finally, the advantages of using integrated simulation techniques for testing delivered implementations with respect to their requirements were explained. Especially, through coupling the simulation with standard unit testing frameworks and continuous integration solutions a seamless build and test execution management throughout the development process is achieved.

As shown so far, the proposed modeling technique works well for requirements management in the context of the academic case study. However, compared to industrial systems the complexity of the case (i.e. the stamping component) is rather limited. Therefore, it is necessary to investigate the suitability of the technique in more complex cases with a more diverse set of requirements (i.e. more complex scenarios and/or more complex declarative functions on the top-level including a variety of constraints). Here, also quality measurements for modeling a more fine-grained assessment of the system behavior should be included. Furthermore, the integration of physics simulation into the test suite is not finished. In particular, adapted build and test execution reports have to be designed such that their appearance resembles the needs of mechatronic systems engineers rather than software engineers only. Finally, the acceptance in practice has to be evaluated (including possible revisions) to ensure its adoption.

5. Summary and Outlook

In this paper a modeling technique for the integration of requirements management in mechatronic development processes was introduced. The applied modeling concepts and terms were outlined and the related tasks and workflows were sketched. Finally, the technique was applied to an academic case study in order to show the feasibility. Pending research tasks address the detailing of the modeling technique regarding the refinement task and its application to an industrial case study. In addition, further research activities will extend the concept to fault-tolerant systems, which includes a method for modeling faults and an automatic generation of fault handling routines within control programs.

Acknowledgement

The authors would like to thank the German Federal Ministry of Economics and Technology (BMW) for the sponsoring of the research project “IMoMeSA” (Funding code: 435 ZN).

References

- [1] Garetti M, Taisch M. *Sustainable manufacturing: trends and research challenges*. In: Production Planning & Control, Vol.23, 2012, pp. 83-104
- [2] Clark K, Wheelwright S. *Managing New Product and Process Development: Text Cases*. The Free Press, New York, 1993
- [3] Russwurm S. *Software: Die Zukunft der Industrie*. In: Sendler U (Hrsg.). *Industrie 4.0 – Beherrschung der industriellen Komplexität mit SysLM*. Springer, Heidelberg, 2013, pp. 21-36
- [4] Eigner M. *Modellbasierte Virtuelle Produktentwicklung auf einer Plattform für System Lifecycle Management*. In: Sendler U (Hrsg.). *Industrie 4.0 – Beherrschung der industriellen Komplexität mit SysLM*. Springer, Heidelberg, 2013, pp. 91-110
- [5] Zaeh MF, Lindworsky A. *Automatic Model Generation for Virtual Commissioning*. In: CIRP: Proceedings International Conference on Competitive Manufacturing, Paris, 2010, pp. 27-32
- [6] Estefan JA. *Survey of Model-Based Systems Engineering (MBSE) Methodologies*. INCOSE MBSE Focus Group 25, 2007
- [7] Schierbaum T, Anacker H, Gausemeier J. *Formalisierte Anforderungen in der Entwicklung mechatronischer Systeme*. In: Maurer M, Schulze SO, Tag des Systems Engineering 2012, Hanser Verlag, München, 2012
- [8] Hofmann HF, Lehner F. *Requirements Engineering as a SuccessFactor in Software Projects*. IEEE software, Vol. 18, No. 4, 2001, pp. 58-66.
- [9] Booch G, Rumbaugh J, Jacobson I. *The Unified Modeling Language User Guide*. Addison-Wesley, 2nd edition, 2005
- [10] Stahl T, Völter M. *Model-Driven Software Development*. John Wiley & Sons Ltd, Chichester, 2006
- [11] Gausemeier J, Dumitrescu R, Steffen D. *Systems Engineering in der industriellen Praxis*. Study of the Heinz Nixdorf Institute and the Fraunhofer IPT – Project Group Mechatronic Systems Design, Paderborn, 2013
- [12] Friedenthal S, Moore A, Steiner R. *A Practical Guide to SysML – The Systems Modeling Language*. Morgan Kaufmann, 2012
- [13] Kallmeyer F. *Eine Methode zur prinzipiellen Lösungen mechatronischer Systeme*. Ph. D. dissertation, University of Paderborn, HNI-Verlagsschriftenreihe 42, Paderborn, 1998
- [14] VDI. *Entwicklungsmethodik für mechatronische Systeme*. Verein Deutscher Ingenieure e.V. (VDI), Beuth Verlag, Berlin, 2004
- [15] Litto M, Korajda I, Mangold C, Angerbauer R, Hils W, Lerche M. *Baukastenbasiertes Engineering mit Föderal*. Verein Deutscher Maschinen- und Anlagenbau, VMDA-Verlag, Frankfurt am Main, 2004
- [16] Litto M. *AQUIMO – Ein Leitfaden für Maschinen- und Anlagenbauer*. Verein Deutscher Maschinen- und Anlagenbau, VMDA-Verlag, Frankfurt am Main, 2010
- [17] Gausemeier J, Frank U, Donoth J, Kahl S. *Specification technique for the description of self-optimizing mechatronic systems*. In: *Research in Engineering Design*. Vol 20, Springer, 2009, pp. 201-223
- [18] Thramboulidis K. *Model Integrated Mechatronics – Toward a New Paradigm in the Development of Manufacturing Systems*. In: IEEE Transactions on Industrial Informatics, Vol. 1, No. 1, 2005, pp. 54-61
- [19] Lüder A, Hundt L, Föhr M, Holm T, Wagner T, Zaddach J. *Manufacturing System Engineering with Mechatronical Units*. Conference on Emerging Technologies and Factory Automation, IEEE 2013, pp. 1-8
- [20] Schedl S. *Integration von Anforderungsmanagement in den mechatronischen Entwicklungsprozess*. Ph. D. dissertation, Technical University of Munich, Utz-Verlag (Forschungsberichte iwb 229), 2008
- [21] Davis A. *Just enough requirements management*. Dorset House Publishing, New York, 2013
- [22] Pohl K, Rupp C. *Basiswissen Requirements Engineering*. dpunkt.verlag, Heidelberg, 2009
- [23] Hood C, Wiebel R. *Optimieren von Requirements Management & Engineering*. Springer, Berlin, 2005.
- [24] Hood C, Wiedemann S, Fichtinger S, Pautz U. *Requirements Management: The Interface Between Requirements Development and All Other System Engineering Processes*. Springer, Berlin, 2008
- [25] Jung C. *Anforderungskklärung in interdisziplinärer Entwicklungsumgebung*. Ph. D. dissertation, Technical University of Munich, Verlag Dr. Hut, 2006
- [26] Reiss T, Schuller J. *Anforderungsmanagement: Der Kernprozess einer Mechatronikentwicklung*. In: Gausemeier J, Rammig F, Schäfer W, 2009, pp. 203-216.
- [27] Gürtler M, Kortler S, Helms B, Berkovich M, Leimeister JM, Krcmar H, Shea K, Lindemann U, Maurer M. *Vom Anforderungslisten zum konzeptionellen Design*. In: Dienstleistungsmodellierung 2012, Springer, 2012, pp. 96-113
- [28] Hölzl F, Feilkas M. *AutoFocus 3: A Scientific Prototype for Model-Based Development of Component-Based, Reactive, Distributive Systems*. In: Model-Based Engineering of Embedded Real-Time Systems, Springer, 2010, pp. 317-322
- [29] Hummel B. *Integrated Behavior Modeling of Space-Intensive Mechatronic Systems*. Ph. D. dissertation, Technical University of Munich, 2011
- [30] Beck K. *JUnit Pocket Guide*. O'Reilly Media, Sebastopol, 2004
- [31] Bullet Physics Library. *JBullet – Java port of Bullet Physics Library*. <http://jbullet.advel.cz> (7th February 2014)